

# 4 Main Database Types



## Relational (SQL)

Structured data in tables with rows and columns, connected by relationships

- Structured Query Language (SQL) based
- ACID compliance (reliable transactions)
- Complex queries with JOINS
- Strong data consistency
- Mature ecosystem & tools
- Standardized SQL language

### Best For:

Financial systems, E-commerce, CRM, ERP, Banking, Inventory management

### Examples:

MySQL, PostgreSQL, Oracle, SQL Server



## NoSQL

Flexible, unstructured data storage designed for scalability and performance

- Horizontal scaling (sharding)
- Flexible schema design
- High performance for simple queries
- Document, key-value, column, graph types
- Built for modern web applications

### Best For:

Real-time web apps, Big data, Content management, IoT

### Examples:

MongoDB, Cassandra, Redis, DynamoDB



## Object-Oriented

Data stored as objects, similar to object-oriented programming concepts

- Complex data relationships
- Inheritance and polymorphism
- Direct object persistence
- Eliminates object-relational mapping
- Rich data modeling capabilities

### Best For:

CAD systems, Multimedia applications, Complex modeling

### Examples:

db4o, ObjectDB, Versant, Gemstone



## Hierarchical

Tree-like structure with parent-child relationships, data flows in one direction

- Fast data retrieval (predefined paths)
- Simple one-to-many relationships
- Efficient for read-heavy operations
- Clear data organization
- Minimal data redundancy

### Best For:

File systems, XML databases, Organizational charts

### Examples:

IBM IMS, Windows Registry, LDAP directories